

DLMAG HW1 REPORT

Release decade and 1980s release market from 30 s of audio

Frozen and fine-tuned MERT-v2, Whisper language ID, zero-shot Qwen2-Audio

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Code, README, requirements.txt: github.com/liu-yu-tung/DLMAG/tree/submission

Checkpoints (8 files, 6.5 GB, SHA-256 listed): github.com/liu-yu-tung/DLMAG/releases/tag/hw1

Outline

- 1 Common to both tasks**
Data and metric, the shared model, how A and B differ, preprocessing, fine-tuning, model selection, MERT-v2 layers, which stem carries the label, Qwen2-Audio prompts slides 3–12
- 2 Task A: release decade**
Model and validation results, confusion matrix and neighbouring-decade errors, Qwen2-Audio on A slides 13–15
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What did not help, errors and limitations, reproducibility, references slides 20–23

Tasks, data and metric

Task A: release decade

6 classes: 1960s, 1970s, 1980s, 1990s, 2000s, 2010s

1,290 clips: train 1,026 (171 per class), validation 132 (22 per class), test 132

Task B: 1980s release market

6 classes: US, UK, Brazil, Spain, Germany, Italy

1,002 clips: train 798 (133 per class), validation 102 (17 per class), test 102

Audio

30 s excerpts, 24 kHz mono WAV.
Nothing above 12 kHz.

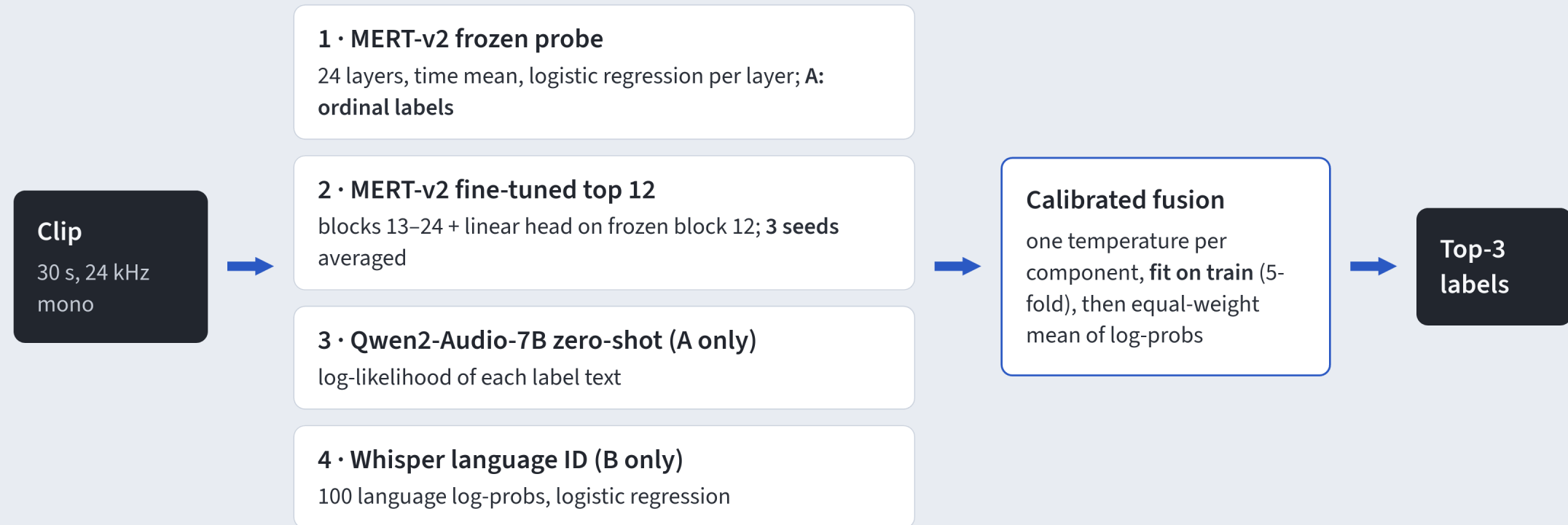
Metric

$S = \text{Top-1} + 0.5 \times \text{Top-3}$. Chance 0.417,
maximum 1.5.

Splits

Train: fitting and every choice.
Validation: a check only. Test: predicted once.

Pipeline



$A = 1 + 3 + 2 \cdot A \text{ fallback (--no-alm, less GPU memory)} = 1 + 2 \cdot B = 1 + 4 + 2$

Models A and B: what is shared, what differs

Shared: frozen MERT-v2 probe + MERT-v2 fine-tuned top 12 (same settings, 3 seeds), one temperature per component fit on train, equal-weight fusion, same selection protocol.

Aspect	Task A: decade	Task B: market
Classes	6 decades, ordered	6 countries, unordered
Training labels	ordinal soft labels (0.1 per neighbouring decade)	plain one-hot labels
Third component	Qwen2-Audio zero-shot label scores	Whisper language ID
MERT-v2 layer sweep	flat: era cues in every layer	market cues in the upper layers
Main error	the neighbouring decade	US and UK; Germany, Spain, Italy into UK
Validation S	1.000 (fallback without Qwen: 0.996)	1.064

Preprocessing and features, every step

Step	What is done
Loading	float32 waveform at the native 24 kHz; no trimming, no loudness normalization (level may carry era cues)
MERT-v2-30s	raw waveform in (its extractor only pads, do_normalize = false), fp32; each of 24 layers gives 750 frames $\times$ 1024, averaged over time
Probe	per layer: StandardScaler + L2 logistic regression ($C = 0.1$); log-probs averaged over the 24 layers. A: each clip also counts with weight 0.1 for each neighbouring decade (ordinal soft labels)
Whisper (B)	resample to 16 kHz; one decoder step after the start token; softmax over the 100 language tokens; StandardScaler + logistic regression
Qwen2-Audio (A)	resample to 16 kHz; LLM in 8-bit; label log-likelihoods minus their mean on train
Fusion	per component, one temperature fitted on its 5-fold out-of-fold log-probs on train; then the equal-weight mean

Fine-tuning the top 12 MERT-v2 blocks

Trained

blocks 13–24 + head (time mean, LayerNorm, dropout 0.2, linear)

Input

the cached block-12 output of frozen MERT-v2 (fp16), one pass per clip

Augmentation (only one)

random 20 s crops in training; the full 30 s at evaluation

Optimizer

AdamW, lr $1e-5$ blocks and $1e-3$ head, weight decay 0.01, 1-epoch warm-up then cosine

Run

15 epochs, batch 8, bf16 autocast; seeds 0, 1, 2, log-probs averaged

Labels

A: ordinal soft labels (0.1 per neighbour). B: plain. All settings fixed before any run.

Model selection without touching validation

5 stratified folds on train (1,026 / 798 clips) give an out-of-fold score for every candidate, compared with the current recipe by a paired bootstrap 95% CI.

Validation is a check only. When we looked at it early it misled twice: it called Whisper language on B noise (5-fold: +0.059) and overstated a stacker on A.

Artist leakage check: no artist field, so proxy groups are Ward clusters of MERT embeddings, kept on one side of each fold. Pessimistic: it also splits off true same-era neighbours.

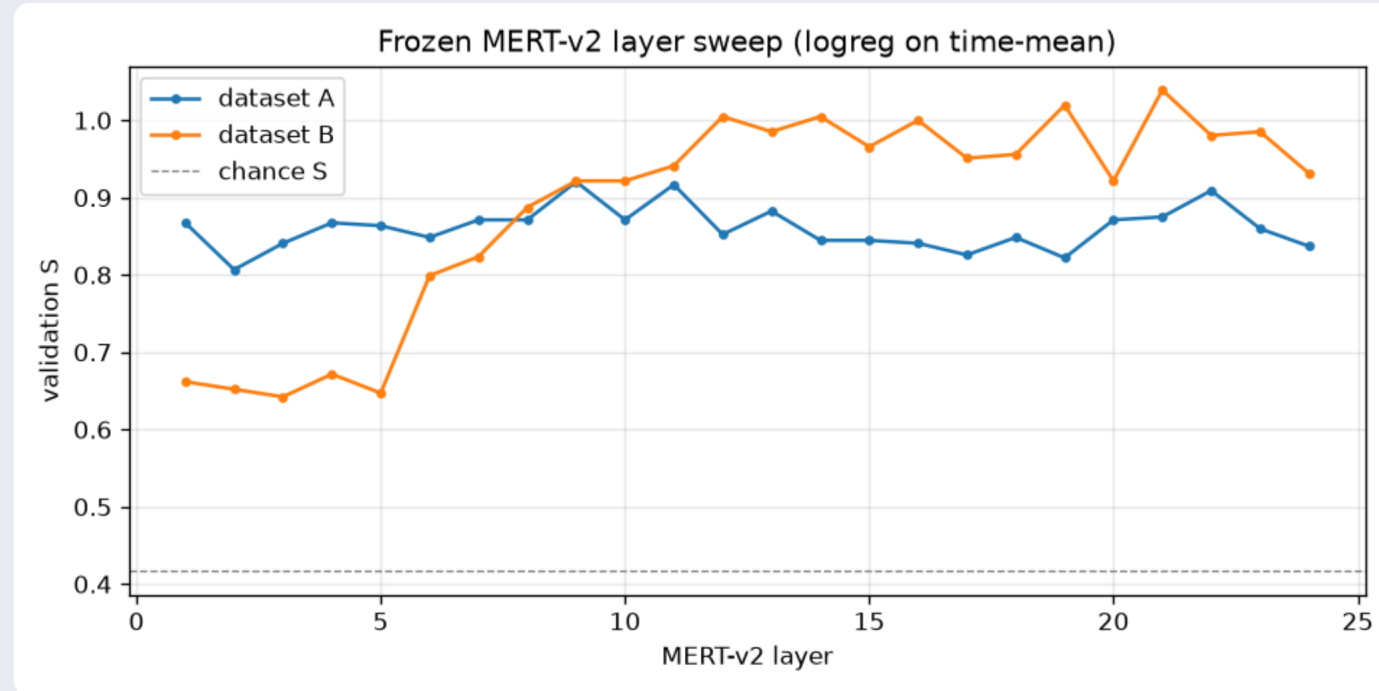
Last change, fine-tuned top 12 added: positive for all 3 seeds on 5-fold and grouped folds.

Recipe	5-fold	grouped	valid.
A	0.969	0.952	1.000
A fallback	0.943	0.932	0.996
B	1.051	1.014	1.064

S of the final recipes. 5-fold and grouped: out-of-fold on train.

Grouped gain from fine-tuning: A +0.038 [+0.018, +0.058], B +0.024 [+0.005, +0.044].

What the MERT-v2 layers know



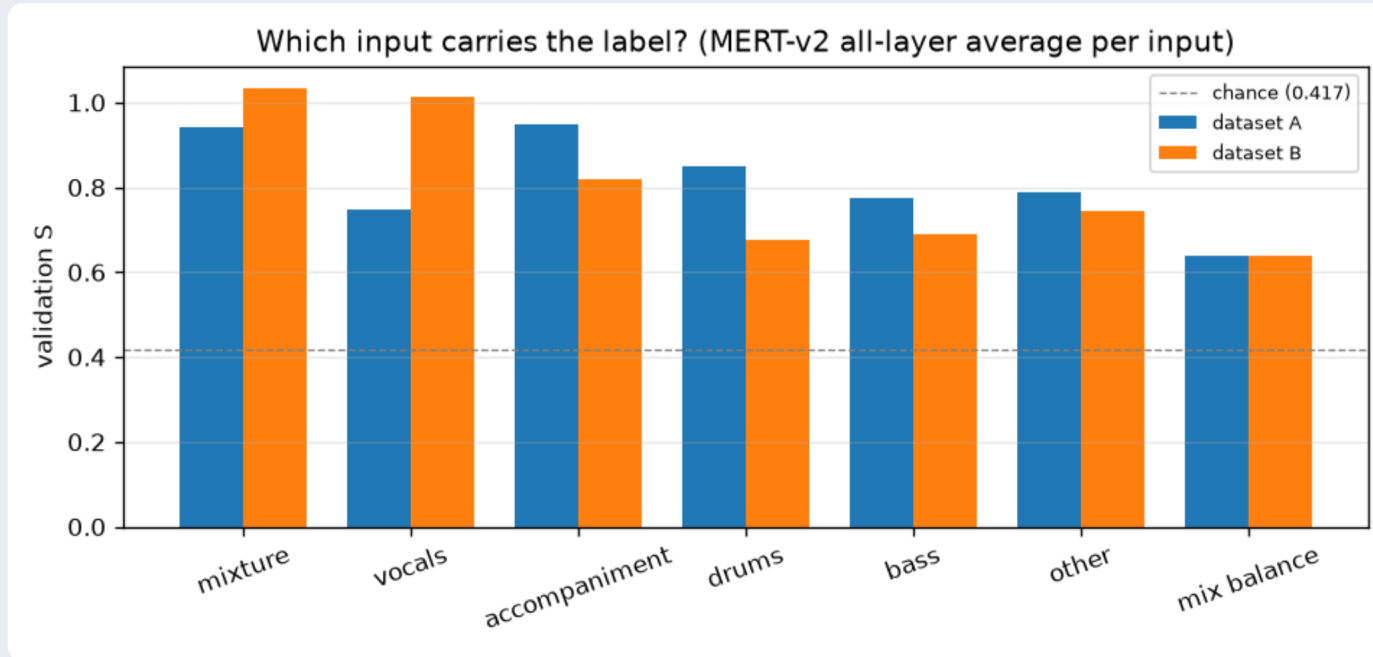
Logistic regression on one layer's time mean; **validation S**.

B: layers 1–5 near 0.65, a climb over 6–11, then 0.92–1.04. Market information sits in the upper layers.

A: flat, 0.81–0.92 at every layer. Era cues are spread through the network.

A best layer picked on 132 clips is optimistic, so we average all 24.

Which part of the mix carries the label



MERT-v2 all-layer probe on each Hybrid Demucs stem (accompaniment = drums + bass + other); validation S.

A: accompaniment 0.947 $\approx$ mixture 0.943, vocals only 0.746. The era is in the backing track: instruments and production.

B: vocals 1.015 $\approx$ mixture 1.034, accompaniment only 0.819. The market is in the singing.

No stem clearly beats the mixture, and stems as extra inputs gave no gain (slide 20), so the models use the mixture.

Fine-tuning depth

Component alone	A 5-fold	A valid.	B 5-fold	B valid.
Frozen probe	0.916	1.011	0.964	1.034
Fine-tune top 4 blocks	0.909 (2 seeds)	0.943	0.996 (1 seed)	0.956
Fine-tune top 8 blocks	0.941	0.974	1.031	1.029
Fine-tune top 12 (used)	0.950	0.992	1.038	1.049

- 3 seeds unless noted, log-probs averaged. The A frozen probe uses ordinal labels.
- Alone, deeper is better on both tasks (5-fold): +0.074 over the probe on B, +0.034 on A.
- Inside the full recipe the gain shrinks, since Qwen (A) and language (B) cover part of it: A +0.022 [+0.001, +0.042], B +0.018 [0.000, +0.036].
- On validation the frozen A probe is the best single component (1.011): 132 clips, inside the CI.

Qwen2-Audio zero-shot: prompts and parsing

Prompt	Text (audio first, chat template)
A plain	In which decade was this song released? Answer with one of: 1960s, 1970s, 1980s, 1990s, 2000s, 2010s.
A cues	adds before it: Listen to the recording and production: audio fidelity, mixing and loudness, drum sound, synthesizers or electric guitars, and vocal style.
B plain	This song was released in the 1980s. In which country was it released? Answer with one of: United States, United Kingdom, Brazil, Spain, Germany, Italy.
B cues	adds after the first sentence: Listen to the language of the lyrics, the accent and singing style, and the musical genre.

Parsing: none needed. For each of the 6 labels we score $\log p(\text{label text} + \text{end-of-turn token} \mid \text{audio, prompt})$ and rank them. Every clip gets a valid, ordered top-3, so there are **0 invalid outputs** by construction.

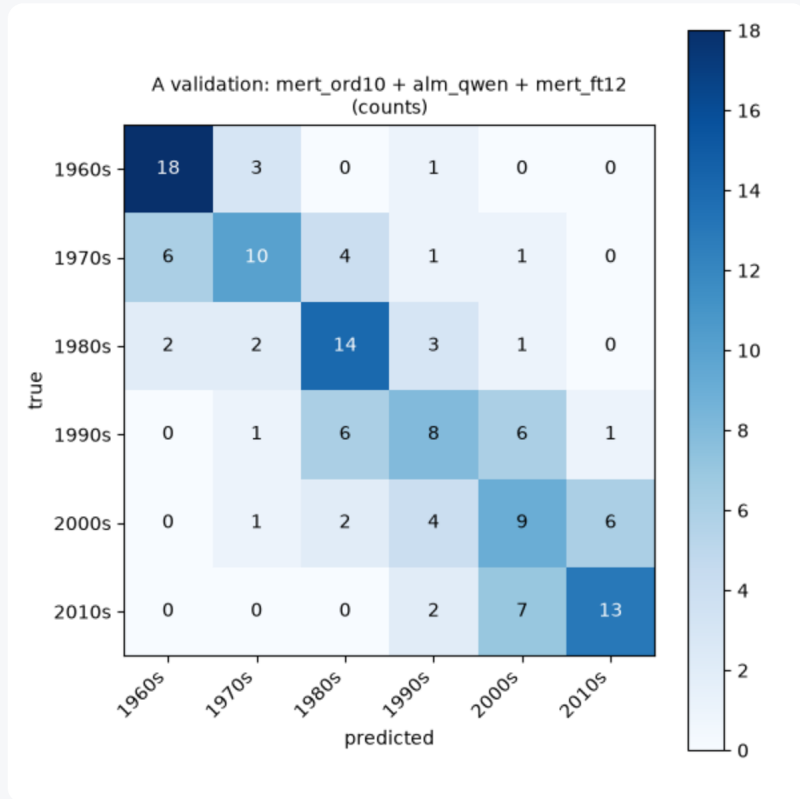
Task A: model and results

Submitted: MERT-v2 probe with ordinal soft labels + zero-shot Qwen2-Audio (train-mean corrected) + fine-tuned top 12, calibrated equal weight.

Model (validation, 132 clips)	Top-1	Top-3	S
submitted	0.545	0.909	1.000
fallback (--no-alm): probe (ordinal) + fine-tuned top 12	0.538	0.917	0.996
probe (ordinal) alone	0.553	0.917	1.011
fine-tuned top 12 alone	0.530	0.924	0.992
Qwen2-Audio, train-mean corrected, alone	0.394	0.773	0.780

Chance S = 0.417. One clip moves Top-1 by 0.008, so **differences under about 0.05 are noise**. The recipe was chosen on **train 5-fold S** (0.969; grouped 0.952).

Task A: most errors are one decade off

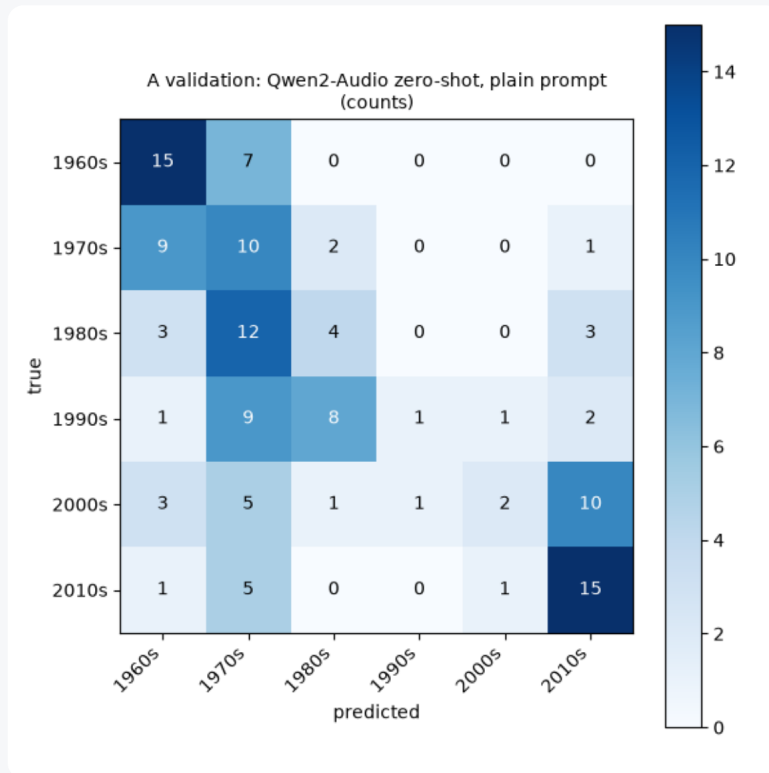


Validation · submitted model · counts · 22 clips per decade

78% of the 60 top-1 errors are one decade off (47). If wrong answers were spread evenly: 36%.

- Two decades off: 10. Three: 3. Four or more: 0.
- Easiest are the ends: 1960s 18/22, 2010s 13/22. Hardest: 1990s 8/22 and 2000s 9/22, which leak to both sides.
- The model learned a time axis: sound changes gradually, so adjacent decades overlap.
- Ordinal soft labels use this (probe alone, 5-fold S): weight 0 gives 0.892, 0.1 gives 0.902, 0.25 gives 0.913, 0.35 gives 0.849. We fixed 0.1 in advance.

Task A: Qwen2-Audio zero-shot



Validation, 132 clips · matrix: plain prompt, counts

Prompt	Top-1	Top-3	S	Most frequent answers
plain	0.356	0.750	0.731	1970s 48, 1960s 32, 2010s 31
cues	0.356	0.720	0.716	1970s 76, 2010s 26

- Almost never answers 1990s (2 clips) or 2000s (4). The cue prompt makes the skew stronger, so we use the plain one.
- Subtracting each label's mean score on train reduces the skew: S 0.731 to 0.780.
- Why keep it: on clips the MERT probe gets wrong, Qwen's top-1 is right 28% of the time (5-fold), the most of any A component.

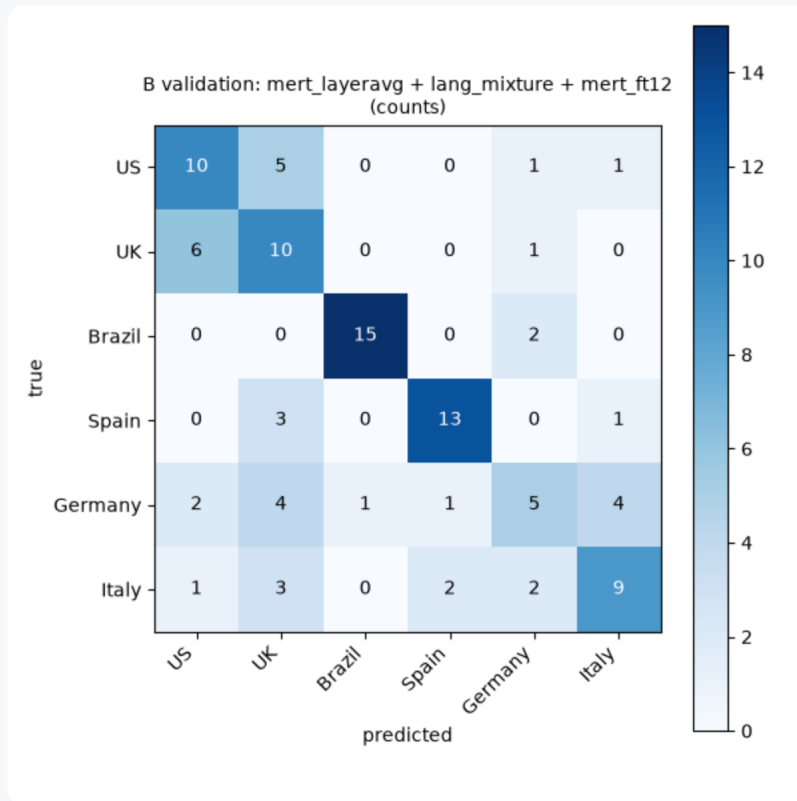
Task B: model and results

Submitted: MERT-v2 probe + Whisper language ID + fine-tuned top 12, calibrated equal weight.

Model (validation, 102 clips)	Top-1	Top-3	S
submitted	0.608	0.912	1.064
probe alone	0.608	0.853	1.034
fine-tuned top 12 alone	0.598	0.902	1.049
Whisper language alone	0.578	0.804	0.980

Chance S = 0.417. One clip moves Top-1 by 0.010, so **differences under about 0.05 are noise**. The recipe was chosen on **train 5-fold S** (1.051; grouped 1.014). Fusion keeps the probe Top-1 and **raises Top-3 (0.853 to 0.912)**.

Task B: errors follow the sung language

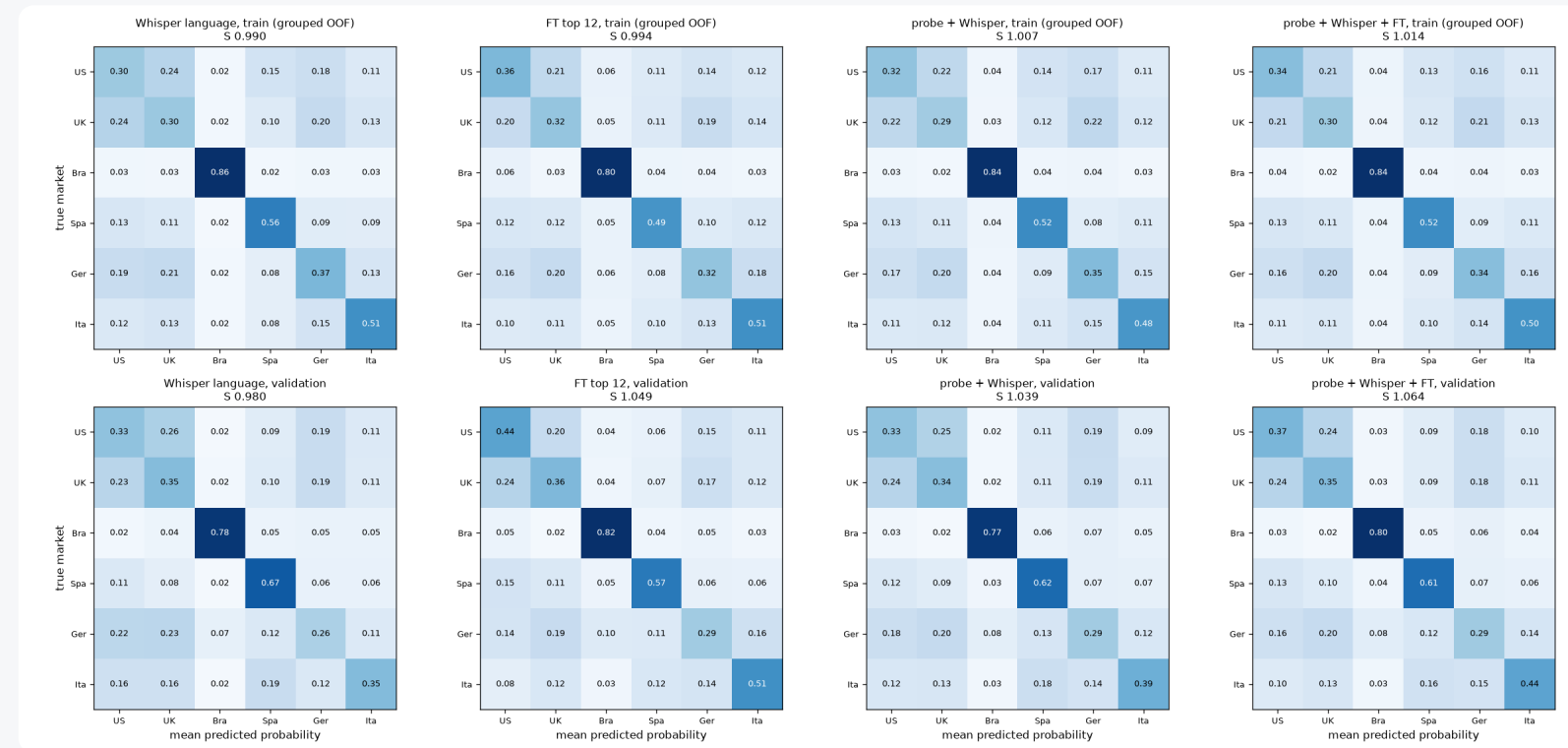


Validation · submitted model · counts · 17 clips per market

11 / 40 top-1 errors are US and UK mixed up.

- Easiest: Brazil 15/17 and Spain 13/17, with Portuguese and Spanish singing.
- Hardest: Germany 5/17, predicted as UK 4, Italy 4, US 2. UK is the most common wrong answer (15 of 40).
- Errors are not between neighbouring countries but between markets with similar sung language, mostly English.
- Market is not the same as language, but the model leans on it.

Task B: the language cue



Row-normalized mean predicted probability. Top row: train, grouped out-of-fold. Bottom row: validation.

0.989

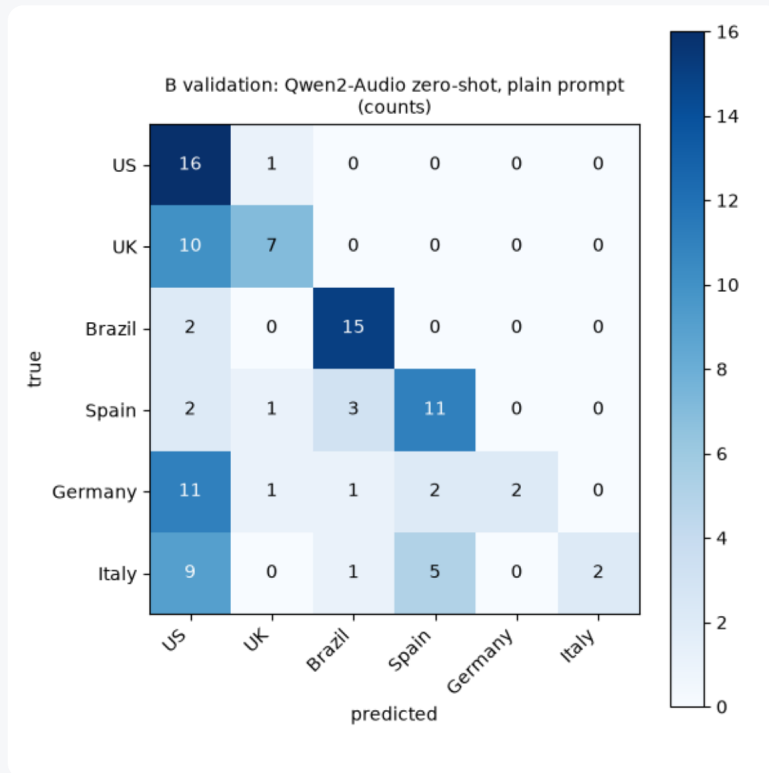
Whisper language alone, 5-fold S (MERT probe: 0.964)

71%

same top-1 answer from Qwen and Whisper (train), so B uses the smaller Whisper

Qwen answers by language (train):
Portuguese → Brazil 117/119, German → Germany 22/24, English → US 384/481

Task B: Qwen2-Audio zero-shot



Validation, 102 clips · matrix: plain prompt, counts

Prompt	Top-1	Top-3	S	Most frequent answers
plain	0.520	0.765	0.902	US 50, Brazil 20, Spain 18
cues	0.441	0.696	0.789	Spain 58, Brazil 19, US 17

- Strong for zero-shot (0.902): it hears the language. Germany and Italy are answered for only 2 clips each; English singing is mapped to the US.
- The cue prompt is worse and pushes answers to Spain, so we use the plain one.
- Not in the B model: its top-1 matches Whisper on 71% of clips, so it mostly repeats the language cue (slide 18).

What did not help

Each model alone, 5-fold out-of-fold S on train	A	B
Hand-crafted librosa features, logistic regression	0.730	0.576
CLAP audio embedding (laion/larger_clap_music)	0.567	0.516
Short-Chunk CNN from scratch on log-mel	0.813	0.657
Hybrid Demucs encoder features	0.852	0.684
MuQ-large	0.867	0.824
MERT-v2 probe, for comparison (A: ordinal)	0.916	0.964

- Also tried, no gain under the selection rule: separated stems (vocals, accompaniment, drums, bass) as inputs, mix-balance and production features, per-class fusion weights, Qwen2-Audio hidden states.
- Chunk length, hand-crafted features on A (validation S, clip score = mean over chunks): 5 s 0.871, 10 s 0.875, 15 s 0.814, 30 s 0.792. The 10 s version was in an earlier A recipe; on top of the final one it adds nothing (grouped +0.006 [−0.012, +0.024]).

Errors and limitations

Limitation	What it means here
Neighbouring decades	the main error on A (78% of errors); 1990s and 2000s are the weakest classes
English-sung markets	the main error on B (US, UK, and many German and Italian releases); the model leans on language, which is not the label
Small validation set	132 / 102 clips, so differences under about 0.05 S are noise; we selected on train cross-validation instead
No artist IDs	train folds may share artists; grouped folds are a proxy and lower S by 0.017 (A) and 0.037 (B)
Input	mono, 24 kHz (no stereo image, nothing above 12 kHz), one 30 s excerpt per song
Cost	submitted A needs Qwen2-Audio (16 GB download, about 13 GB GPU memory); the fallback (about 6 GB) scores 0.996 vs 1.000
Ported models	MERT-v2 and MuQ needed load-time patches on transformers 5; checked against repeated loads, not the authors' setup

Reproducibility

- **Release:** code, README and pinned requirements.txt (Python 3.12, torch 2.14.0+cu130, transformers 5.17.0); checkpoints with SHA-256 (links on page 1).
- **Check:** a fresh venv built only from requirements.txt, run from audio, gives the same top-3 lists (order included) as the training-time features on 234/234 test clips, in both modes. A repeated run is byte-identical.
- **Runtime** (RTX 4060 Ti 16 GB, 234 test clips): 7 min 3 s default, 1 min 46 s with --no-alm. Seeds 0, 1, 2; settings fixed before training.

Bug 1, found and fixed

MERT-v2's rotary frequencies are a non-persistent buffer. Loaded on the meta device by transformers 5, they were left uninitialized, so features changed between loads (25–55%). Fix: restore the buffer after loading.

Bug 2, found and fixed

At inference the fine-tuned blocks reused a rotary table built in fp32; training built it under bf16 autocast. Log-probs moved by up to 0.81. Fix: clear the cache, so inference matches training exactly.

References

Pretrained models

MERT-v2-30s. Y. Li et al., MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training, ICLR 2024.

huggingface.co/m-a-p/MERT-v2-30s

Qwen2-Audio-7B-Instruct. Y. Chu et al., Qwen2-Audio Technical Report, arXiv:2407.10759, 2024.

huggingface.co/Qwen/Qwen2-Audio-7B-Instruct

Whisper large-v3-turbo. A. Radford et al., Robust Speech Recognition via Large-Scale Weak Supervision, ICML 2023.

huggingface.co/openai/whisper-large-v3-turbo

CLAP. Y. Wu et al., Large-scale Contrastive Language-Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation, ICASSP 2023.

huggingface.co/laion/larger_clap_music

MuQ-large. H. Zhu et al., MuQ: Self-Supervised Music Representation Learning with Mel Residual Vector Quantization, arXiv:2501.01108, 2025.

huggingface.co/OpenMuQ/MuQ-large-msd-iter

Hybrid Demucs. A. Défossez, Hybrid Spectrogram and Waveform Source Separation, ISMIR 2021 MDX Workshop.

[torchaudio HDEMUCS_HIGH_MUSDB_PLUS](https://torchaudio.github.io/HDEMUCS_HIGH_MUSDB_PLUS)

Methods

M. Won, A. Ferraro, D. Bogdanov, X. Serra, Evaluation of CNN-based Automatic Music Tagging Models, SMC 2020 (**Short-Chunk CNN**).

C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On Calibration of Modern Neural Networks, ICML 2017 (**temperature scaling**).

Code

PyTorch and torchaudio; Hugging Face Transformers; bitsandbytes; scikit-learn (F. Pedregosa et al., JMLR 2011); librosa (B. McFee et al., SciPy 2015); NumPy, SciPy, pandas, soundfile, matplotlib. Course: DLMAG lecture slides and the HW1 specification.